LESSON 119 (0.5 GROUND and 1.5 DUAL, STAGECHECK MOCK)

Lesson Objective

Review all basic maneuvers with concentration on preparing the student for the first stage check and solo flight.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Ground)

Review as Applicable

- Ask the student if they have any questions before going on the flight.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
    - Personal minimums.
    - FIT minimums.
    - Aircraft.
    - Airspace.
- Review the students planning with a focus on:
    - Weather, weight and balance, performance, flight planning, and fuel calculations.
    - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
    - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
    - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.
- Walk around the airplane and quiz the student on:
    - Antennas.
    - Aircraft structure.
    - Flight controls.
    - Systems.
    - Placards and markings.

Cockpit Management

- Ensure the student came prepared with all of the necessary documents and equipment (kneeboard, headset, airport diagram, checklists, etc….)

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Engine Starting

- Verify correct procedures.

Radio Communications

- Ensure that the student can adjust their headset volume, intercom volume, squelch, and comm volumes.
- Ensure that the student can set the right frequencies.

Use of Checklists

- Ensure that the student is using the correct checklist and verbalizing all items.
    - No items should be skipped.
- Ensure that the student is performing the correct action for "Read and Do" vs "Do and Verify" checklists.

Taxiing

- Verify correct procedures for the taxi check.
- Taxi speeds should be at a brisk walking pace (5kts GS).
- Power to idle before braking.
    - No riding the breaks.
- Taxi instructions <u>must</u> be briefed.

Normal and Crosswind Takeoffs

- Ensure runway line-up check is performed.
- Students must look at the windsock prior to establishing a crosswind correction (ATIS winds might not be what the windsock is showing).
- Verify proper rudder usage.
- Verify rotation happens smoothly at 55 knots.
- Crosswind correction should be kept up until rotation.
- On the departure leg, the student should track the runway depending on the winds.

Emergency Operations Including Emergency Approach and Landing

- Engine failure in flight:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.

        - Trimming for V_{glide}.
        - Wire strike avoidance.
        - Choice of landing point.
        - Checklist usage.
        - Verify recovery happens before reaching 500 ft.

Slow Flight

- <u>Student says, student does.</u>
    - Verify correct procedure.
- Points of Emphasis.
    - Pre-maneuver checklist.
    - Trim.
    - Bank angle (in turns).
    - Smooth control.
    - Pitch for airspeed and power for altitude.
- Standards:
    - Altitude: +/- 150ft.
    - Heading: +- 20 degrees.
    - Airspeed: +/-10 knots.

Stalls

- Power-off:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Pre-maneuver checklist.
        - Smooth control.
        - Establish and maintain approach speed.
        - Announce imminent stall.
        - Coordination.
    - Standards:
        - Heading: +/-30 degrees.
        - Altitude loss during recovery: not exceeding 250 ft.
- Power-on:
    - This can either be done in the short-field takeoff or climb configuration.
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Pre-maneuver checklist.
        - Smooth control.
        - Altitude must be held until reaching rotation speed.

        - Announce imminent stall.
        - Coordination.
    - Standards:
        - Heading: +/-30 degrees.
        - Altitude loss during recovery: not exceeding 250 ft.

Ground Reference Maneuvers

- Turns around a point:
    - Student says, student does.
        - Verify correct procedure.
    - Points of Emphasis.
        - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
        - Bank variation.
        - Choosing an adequate and safe visual reference point.
    - Standards:
        - Bank: maximum 45 degrees.
        - Altitude: +/- 150 ft.
        - Airspeed: +/- 10 knots.
- S-turns:
    - Student says, student does.
        - Verify correct procedure.
    - Points of Emphasis.
        - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
        - Bank variation.
        - Choosing an adequate and safe visual reference point.
    - Standards:
        - Bank: maximum 45 degrees.
        - Altitude: +/- 150 ft.
        - Airspeed: +/- 10 knots.
- Rectangular Course:
    - Student says, student does.
        - Verify correct procedure.
    - Points of Emphasis.
        - Entry procedures: 45 degrees to the downwind, sufficient space to set up, and proper spacing from point.
        - Bank variation.
        - Choosing adequate and safe visual reference points.
    - Standards:
        - Bank: maximum 45 degrees.

        - Altitude: +/- 150 ft.
        - Airspeed: +/- 10 knots.

Traffic Patterns

- Verify altitude is within 150ft and airspeed is within +5/-0 knots.
- Correct for wind drift.
- Verify correct airspeeds throughout the pattern.

Rejected Takeoff and Engine Failure after Takeoff

- Verify correct procedures.
- This must be briefed during the takeoff briefing.
    - If the engine fails on the runway, the student must maintain directional control and stop on the runway (simulated).
    - If the engine fails after rotation and sufficient runway is remaining, the student must maintain a safe airspeed and land on the remaining runway (simulated).
    - If the engine fails after rotation and there is insufficient runway remaining, the student must maintain a safe airspeed and choose a place to land (simulated).
    - If the engine fails and there is sufficient altitude, the student must turn back to the airport and land on one of the runways (simulated).
- Evaluate the student's judgment.

Normal, Crosswind, No Flap, and Power-Off Approach and Landing

- Normal/crosswind landing:
    - Verify correct approach speed.
    - The student must conduct a landing where the main touch-down before the nose wheel.
    - At least one of the main wheels should be on the centerline.
    - Ensure that the student is maintaining the proper wind correction.
    - The student must be able to judge when to round out and flare.
- No flap landing:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Normal pattern vs wider pattern.
        - Judgment (different angle of attack, different sight picture, etc.)
        - Airspeeds.
- Power loss on the downwind:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Trimming for V_{glide}.
        - Must land on the first ⅓ of the runway.

Go-Around/Rejected Landing

- If the student has an unstable approach, see if they decide to initiate a go-around.
- If all of the approaches are stable, have the student perform a go-around due to “something” on the runway (airplane, animal, vehicle, etc.)
- Verify correct procedures.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 120.
    - Stage check.
- Assign homework:
    - Review for the stage check!

Completion Standards

This lesson is complete when the student is able to:

1. Operate within the FIT Flight Operations Manual and applicable federal aviation regulations.
2. Perform all maneuvers and procedures to meet or exceed the standards outlined in the Stage 1 completion standards (see pg. 6).
3. Act as pilot in command on a local solo flight.